

LOW ACRYLONITRILE NBR: PATENT LANDSCAPE FOR COMPOSITION, POLYMERIZATION, AND DOWNSTREAM PROCESSING

1. ABSTRACT

This report synthesizes a patent landscape relevant to low acrylonitrile nitrile butadiene rubber (NBR), interpreted as NBR compositions and processes in which acrylonitrile is present at the lower end of the conventional NBR range or is used in formulations that explicitly include low-ACN rubber grades. Across the retained patents, the evidence spans three main themes: (1) direct NBR latex synthesis and monomer-ratio control, (2) latex post-treatment, coagulation, and compounding for dip-molded or reinforcement applications, and (3) downstream use of NBR in masterbatches, dipping compositions, and adhesion systems. The most compositionally relevant disclosures are EP1478672B1 and US11236218B2, which both define NBR-type latices with butadiene-rich and nitrile-containing monomer balances; EP1478672B1 specifies 50 to 89.5 parts diene, 10 to 40 parts nitrile, 0.5 to 10 parts acid monomer, and staged monomer addition to control latex properties, while US11236218B2 describes carboxyl group-containing nitrile rubber containing 56 to 78 wt.% diene units, 20 to 40 wt.% nitrile units, and 2 to 6.5 wt.% carboxylic acid units, prepared by emulsion polymerization at 0 to 25 °C with 0.15 to 0.95 parts chain-transfer agent per 100 parts monomer. Other patents emphasize processing rather than synthesis: US5112910A teaches co-coagulation of aqueous emulsions at 20 to 100 °C to make free-flowing powder mixtures; US3006872A discloses heat-sensitized coagulation of polymer dispersions using poly(ethylene oxide); US12077640B2 uses coagulation and drying to make polysaccharide-elastomer masterbatches; EP3935214B1 formulates RF-free dipping compositions with rubber latex, blocked isocyanate, wax, epoxy, and carboxylic polymers; and US11131058B2 uses carboxy-modified latex in cord treatment with resorcin-formaldehyde and polycarbodiimide. Overall, the landscape shows that low-ACN NBR is most often managed through monomer feed balance, staged addition, low-temperature emulsion polymerization, and post-polymerization coagulation or latex compounding to tune softness, tensile strength, adhesion, and processability.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US5112910A
- Patent Title: Mixtures of thermoplastic polymers in powder form
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 1992
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

Patent 2

- Patent Number: US10336853B2
- Patent Title: Polymer, process and composition
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 3

- Patent Number: US3006872A
- Patent Title: Coagulation of dispersed polymeric organic material with poly(ethylene oxide), ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 1961
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

Patent 4

- Patent Number: EP1478672B1
- Patent Title: Dip-forming latex, dip-forming composition and dip-formed article
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

- Patent Number: US8609877B2
- Patent Title: Cyclic diol-derived blocked mercaptofunctional silane compositions
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

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- Patent Number: US12077640B2
- Patent Title: Polysaccharide-elastomer masterbatch compositions
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2024
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

Patent 7

- Patent Number: US11236218B2
- Patent Title: Latex composition and film molded body
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2022
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

Patent 8

- Patent Number: US11262338B2
- Patent Title: Modified resins and uses thereof
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2022
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

Patent 9

- Patent Number: EP3935214B1
- Patent Title: Aqueous dipping composition
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2023
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

Patent 10

- Patent Number: US11131058B2
- Patent Title: Method for manufacturing core wire for transmission belt, treatment agent, and ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- EP1478672B1 and US11236218B2 are the most directly relevant NBR synthesis disclosures: both use butadiene-rich, nitrile-containing monomer systems, but EP1478672B1 allows 10 to 40 parts nitrile and 0.5 to 10 parts acid monomer with 50 to 89.5 parts diene, while

US11236218B2 defines a narrower carboxylated nitrile rubber window of 56 to 78 wt.% diene, 20 to 40 wt.% nitrile, and 2 to 6.5 wt.% carboxylic acid units.

- EP1478672B1 emphasizes staged monomer feed: initiation with at least 80% of diene, at least 50% of nitrile, and 30 to 85% of acid monomer, then delayed addition of the remaining acid at 20 to 80% conversion; US11236218B2 instead emphasizes low-temperature emulsion polymerization at 0 to 25 °C with 0.15 to 0.95 parts chain-transfer agent per 100 parts monomer.
- US5112910A is not a synthesis patent for NBR itself, but it is relevant to NBR latex handling: it co-coagulates aqueous emulsions of rubber-like copolymers containing up to 40 wt.% acrylonitrile with thermoplastic resins at 20 to 100 °C, showing an industrial route for powder isolation and blending.
- US3006872A and US12077640B2 both center on coagulation of dispersed polymer systems, but US3006872A uses poly(ethylene oxide) as a heat-sensitizer for latex coagulation, whereas US12077640B2 uses inorganic salt or acid coagulation to combine rubber latex with polysaccharide dispersions into masterbatches.
- EP3935214B1 and US11131058B2 are downstream application patents rather than NBR synthesis patents: EP3935214B1 uses nitrile butadiene rubber latex as one option in RF-free textile dipping compositions, while US11131058B2 specifically names carboxy-modified acrylonitrile-butadiene latex in cord treatment with resorcin-formaldehyde and polycarbodiimide.
- US10336853B2 and US11262338B2 are only indirectly relevant: they focus on solution-polymerized itaconate copolymers and modified thermoplastic hydrocarbon resins, respectively, but they illustrate broader formulation trends toward lower VOC, lower oligomer content, and reduced reliance on traditional resin systems rather than NBR synthesis itself.
- Across the primary patents, the recurring process logic is emulsion polymerization for NBR formation, followed by coagulation or latex blending for end-use performance; the newer patents place more emphasis on low-temperature control, chain-transfer management, and formaldehyde-free or low-emission downstream formulations.
- Historical disclosures such as US3006872A and US5112910A focus on coagulation mechanics and powder recovery, whereas newer disclosures such as EP1478672B1, US11236218B2, EP3935214B1, and US11131058B2 focus on property-balanced latexes for gloves, cords, and dipped articles.

4. CONCLUSION

The evidence indicates that low acrylonitrile NBR is best synthesized and processed as a butadiene-rich emulsion copolymer with nitrile content kept in the lower-to-mid conventional NBR range and, where needed, a small carboxylic acid comonomer fraction for adhesion and dip-forming performance. The strongest synthesis guidance comes from EP1478672B1 and US11236218B2: staged monomer addition, low-temperature emulsion polymerization, and explicit chain-transfer control are the most clearly supported levers for tuning latex properties. For downstream use, the patents favor coagulation-based isolation or direct latex compounding, with

modern formulations increasingly avoiding resorcinol-formaldehyde where possible and using blocked isocyanates, epoxy compounds, metal salts, or carboxyl-functional additives to improve bonding and storage stability. Based on the evidence, a practical target window for low-ACN NBR would be butadiene-rich composition with acrylonitrile at about 10 to 40 parts by weight or 20 to 40 wt.% in carboxylated grades, optional acid monomer at about 0.5 to 10 parts by weight or 2 to 6.5 wt.% in functionalized grades, emulsion polymerization at 0 to 25 °C when molecular-weight control is important, and chain-transfer agent use in the low sub-1 phr range where disclosed. For processing, coagulation or co-coagulation at moderate temperature remains the dominant isolation route, while dipped-article and cord applications benefit from carboxylated or otherwise functionalized NBR latexes rather than unmodified low-ACN rubber alone.

5. REFERENCES

- US5112910A | Mixtures of thermoplastic polymers in powder form | Not disclosed | US | 1992 | <https://patents.google.com/patent/US5112910A/en>
- US10336853B2 | Polymer, process and composition | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10336853B2/en>
- US3006872A | Coagulation of dispersed polymeric organic material with poly(ethylene oxide), ... | Not disclosed | US | 1961 | <https://patents.google.com/patent/US3006872A/en>
- EP1478672B1 | Dip-forming latex, dip-forming composition and dip-formed article | Not disclosed | EP | 2015 | <https://patents.google.com/patent/EP1478672B1/en>
- US8609877B2 | Cyclic diol-derived blocked mercaptofunctional silane compositions | Not disclosed | US | 2013 | <https://patents.google.com/patent/US8609877B2/en>
- US12077640B2 | Polysaccharide-elastomer masterbatch compositions | Not disclosed | US | 2024 | <https://patents.google.com/patent/US12077640B2/en>
- US11236218B2 | Latex composition and film molded body | Not disclosed | US | 2022 | <https://patents.google.com/patent/US11236218B2/en>
- US11262338B2 | Modified resins and uses thereof | Not disclosed | US | 2022 | <https://patents.google.com/patent/US11262338B2/en>
- EP3935214B1 | Aqueous dipping composition | Not disclosed | EP | 2023 | <https://patents.google.com/patent/EP3935214B1/en>
- US11131058B2 | Method for manufacturing core wire for transmission belt, treatment agent, and ... | Not disclosed | US | 2021 | <https://patents.google.com/patent/US11131058B2/en>